

Siddhi Kabadi

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EDUCATION

Brown University | M.S. Cybersecurity (CS Track)

May 2028

Santa Clara University | B.S. Computer Science (Cybersecurity Emphasis)

June 2026

- Relevant Coursework: Operating Systems; Computer Security; Algorithms, Object-Oriented Programming, Data Structures (C++); Cryptography; Embedded Systems (ARM); Web Usability and Infrastructure (XML), Calculus I-IV; Discrete Math; Applied Cryptography

PUBLICATIONS & PRESENTATIONS

Aqlan, F., Kabadi, S., *Design Principles and Usability Analysis of Virtual Reality Manufacturing Digital Twins*: MDPI Systems Journal, 2025 (In preparation to submit)

Kabadi, S., Houshmand, S., *ML Framework for Password Policies*: Poster accepted to ACMW Computing Conference

Kabadi, S., Shaghaghi, N., *Hydration Automation: LoRa-based Smart Irrigation for Community Farms*: Poster presented at IEEE MIT URTC, IEEE GHTC

EXPERIENCE

NSF Research Assistant: VR Development and Data Analysis

May 2025 - Present

University of Louisville REU (NSF-funded) | Louisville, KY

- Developed a virtual reality environment in Unity using Photon Fusion to simulate a digital twin system, and collected eye-tracking data and user-interaction events to study task performance within the environment
- Implemented TLS-encrypted server connections, JWT authentication, and role-based access control to protect biometric information, and maintain the integrity of data access pathways. Engineered Python-based data analysis scripts (Pandas, Seaborn, Matplotlib) to process user-interaction logs
- Applied statistical methods (t-tests and Spearman correlation) to evaluate relationships between usability perception (SUS), cognitive load (SIM-TLX) and behavioral data. Proposed data-driven design improvements through a journal paper (2nd author– MDPI Systems Journal) and a poster (IMECE 2025)

Student Researcher in Cybersecurity: Machine Learning and Password Security

December 2024 - March 2025

Santa Clara University | Santa Clara, CA

- Conducted a literature review and comparative analysis of existing password-strength frameworks (zxcvbn and LUDS), to study how deterministic rules and pattern-matching approaches overlook human cognitive habits and bias
- Utilized NLP techniques and trained a BERT model on large real-world password datasets to examine the linguistic and contextual patterns present in manually created credentials
- Fine-tuned a pre-trained BERT (bert-base-cased) model on 470K labeled passwords to classify strength levels (weak, medium, strong), achieving 87% accuracy and 0.81 F1 score on test data. Implemented model training, tokenization, and evaluation using PyTorch and scikit-learn to optimize learning rates, batching, and GPU performance
- Performed correlation analysis and K-Means clustering to evaluate the alignment between my BERT model's predictions and existing security approaches, identifying where modern meters (e.g., zxcvbn) differ structurally.

Student Researcher in Internet of Things

May 2023 - Present

Ethical, Pragmatic, and Intelligent Computing Laboratory | Santa Clara, CA

- Developed a RESTful API using Node.js and Express to log IoT sensor data (moisture, humidity, temperature) into a SQL database
- Created a web-based dashboard using React and JavaScript to visualize real-time soil moisture data from LoRa sensors
- Collaborated with farmers at Veggielution; conducted meetings to refine the dashboard's usability; ensured the system aligned with the farm's needs

Teaching Assistant: Computer Science and Engineering

January 2025 - Present

Santa Clara University | Santa Clara, CA

- Graded assignments and projects for 50 undergraduate students in Introduction to Computer Science. Facilitated weekly lab sessions for students in Introduction to Design Thinking, creating problem framing, ideation, and prototyping exercises
- Mentored students to help understand Python fundamentals (conditionals, recursion, loops); conducted discussions on the social impact of computing

PROJECTS

Task Tracker | SQL, PHP, SQLite, JavaScript, HTML/CSS

- Created a web application with user authentication, task management, and progress tracking abilities using HTML, CSS, JavaScript, and SQL
- Integrated security measures and data integrity with hashed passwords, session management, and a SQLite Database

Click Cleverly | Privacy Engineering, UX Research, Figma

- Conducted privacy-focused research on Terms of Service agreements; analyzed how legal complexity and interface design undermine informed user consent
- Identified and categorized privacy risks including data tracking, third-party sharing, and content ownership through literature review and user studies
- Designed a Figma prototype that models user interaction with privacy policies and explored design approaches to help users better understand privacy risks

TECHNICAL SKILLS

Programming Languages: C/C++/C#, Python, Java/JavaScript, HTML/CSS, SQL, PHP

Software Tools: Tableau, Power BI, Jira, Confluence, Ubuntu, Docker, Unity, Unreal Engine, Postman, Wireshark, John the Ripper, Hashcat

Frameworks: React, Node.js, Express.js, Django, Scikit-learn, Transformers Library, PyTorch, Pandas, Matplotlib, Seaborn